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RESEARCH ARTICLE

Pixel-Based Change Detection in Moving-Camera Videos Using Twin Convolutional Features on a Data-Constrained Scenario

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ABSTRACT This work proposes the Pixel-Based Change Detection using a Moving Camera (PBCD-MC) algorithm for detecting anomalies in a cluttered industrial site. The surveillance of industrial environments may be challenging for an anomaly detection problem, as the frequent presence of pipes, cables, and heavy machinery in the scene can create occlusion, undesirable shadows, metallic reflections, etc. A moving camera, surveilling the environment based on reference videos, may also face parallax differences, tracking misalignments, and illumination changes throughout the day. To address these moving camera surveillance problems, the PBCD-MC method was developed based on a hybrid ensemble feature extractor, combining deep learning methods, responsible for generating high-level features, and tree-based algorithms, responsible for selecting and combining the deep features. In addition, in order to deal with the challenges of training with high-dimensional data, a complexity-reducing training strategy that eliminates the need for large training sets while maintaining a robust detection performance was proposed. The PBCD-MC is assessed using a newly developed pixel-level annotated dataset (VDAO-AS) and is shown to outperform state-of-the-art models, achieving overall $MCC = 0.877$ and overall $DIS = 0.092$ on a pixel-level detection, results that are significantly better than those of its competitors. Also, a series of ablation studies proves the importance of the adopted modules for the achieved performance. Software, datasets, results, and the pretrained weights of the proposed architecture are available at https://github.com/lgtavares/VDAO_Pixel.

INDEX TERMS Convolutional neural networks, industrial environment, moving camera, object detection, tree-based classifiers.

I. INTRODUCTION

Inspecting surveillance videos for change detection (including object placement or removal, detection of moving entities such as people or vehicles, etc.) has gradually become automated by computer-vision frameworks. The abundance

of videos in such applications allows for constructing large datasets suitable for training complex machine learning algorithms, which may achieve high-performance levels, matching and sometimes surpassing human performance.

The ability to fast and precisely detect abnormal events is essential to obtain a quick response, particularly in the context of industrial surveillance, to prevent hazardous events that can damage structures and machinery or endanger workers' safety and health.

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However, the analysis of industrial videos faces problems related to cluttered environments, as the videos often contain many elements that can disrupt the detection process, such as pipes, cables, machinery, and so on. An even more challenging scenario is related to surveillance videos acquired with moving cameras, which may originate from cameras mounted on robots or drones in order to reach wider zones that are dangerous or hardly accessible to humans. In addition to the added complexity of processing such videos, their availability in public datasets is somewhat limited, hindering the training capacity of deep-learning solutions for this particular problem.

It is well known that the most successful video change- and anomaly-detection algorithms include several thousand (or even millions) of trainable parameters [1]. To take advantage of the promising results yielded by deep-learning algorithms in a data-constrained case like the one mentioned above, some restrictions must be placed on the algorithms' framework and learning strategy. This work proposes the Pixel-Based Change Detection using a Moving Camera (PBCD-MC) algorithm, which is a method for detecting changes in an industrial environment from moving-camera videos.

The PBCD-MC not only addresses the limitations of data-constrained scenarios but also introduces a framework optimized for the complexities of industrial environments surveilled by a moving camera, such as illumination changes, camera tremors, parallax and occlusions. By focusing on pixel-level detection, PBCD-MC provides finer granularity and more precise identification of changes, aiming to reduce such unwanted effects on overall detection.

The PBCD-MC algorithm works with two video sequences: one is the reference video, which is acquired in a controlled certified state; the other is the target video, which is acquired on inference time and is where changes and anomalies shall be detected following proper comparison to the reference video.

The proposed PBCD-MC algorithm includes five modules that perform video alignment, feature extraction, data sampling, classification, and post-processing. The alignment module minimises the root-mean-square difference to find the best alignment between reference and target frames. The feature-extraction stage is based on a pre-trained siamese ResNET-50 network [2] with shared weights. In this step, the outputs of both networks (containing the features extracted from aligned reference and target frames) are merged into a single tensor fed to the subsequent processing stage. The classification module is composed of a light gradient boosting machine (LightGBM) [3] with a reduced number of learnable parameters, that allows for training in a limited data-availability scenario. Finally, the post-processing module performs several morphological operations to improve output consistency and overall system performance. The data-sampling module, only used during training, is responsible for downsampling the training data, so that the model can be designed with a smaller number of parameters that are still representative of each class.

The contributions of this work can be listed as:

A. MAIN CONTRIBUTIONS

- **Development of a new dataset, annotated frame-by-frame at the pixel level**, for change detection in videos collected by a moving camera in an industrial environment. In this new dataset, changes are identified by their respective silhouettes, as opposed to the bounding boxes employed in previous works;
- **A pixel-based methodology** for the change-detection classifier that enables the development of a comprehensive and balanced training set from a limited number of frames in the newly developed video dataset;
- **The use of an uncertainty zone** in the annotation process, so that the pixels marked as background/foreground pixels are only the ones for which there is no doubt they are background/foreground. The pixels for which there is some doubt relative to belonging to the background or foreground are marked as belonging to the uncertainty zone;
- **A balanced data reduction module** for the training stage that randomly selects feature tensor samples, reducing the training computational resources usage;
- **A comprehensive experimental assessment** comparing the proposed method to state-of-the-art algorithms.

B. ADDITIONAL CONTRIBUTIONS

- A modular framework for the change-detection algorithm that performs the feature extraction using a deep neural network that has not been trained specifically for the problem at hand, followed by a classification module that benefits from the tensor composed of both the target and reference features, with a reduced number of trainable parameters;
- A post-processing module designed to reduce the incidence of false positives and false negatives, by using morphological operations and a temporal voting window designed to exploit the inter-frame dependency to enhance detection;
- Presentation of extensive ablation studies that demonstrate the efficacy and importance of all separate modules of the proposed change-detection algorithm and validate the proposed pixel-based training methodology;

All these ideas are organized in this paper as follows: the current literature related to this work is reviewed in Section II and the related state-of-the-art gaps are identified; Section III introduces the proposed method, including all its modules and the developed methodology; Section IV details and discusses the training strategies; the experimental assessment, which encompasses ablation studies and comparison to state-of-the-art methods, is presented in Section V. Finally, the conclusions and final remarks are presented in Section VI.

II. RELATED WORKS

Change detection in surveillance videos captured by static cameras has been extensively studied in the literature [4], with

a primary focus on detecting moving objects such as vehicles and pedestrians [5]. A fundamental technique in this domain is background subtraction [6], which exploits knowledge of the static background to identify newly introduced objects, whether stationary or moving. This approach has been formalized in various frameworks, including those addressing challenges like illumination changes and dynamic backgrounds [7]. While traditional background subtraction assumes a fixed camera, recent work has adapted the method to handle moving-camera scenarios [8]. Key to these adaptations is estimating geometric transformation of the background caused by camera motion, enabling robust change detection [9]. Further extensions, such as the work by [10], generalize background subtraction to freely moving cameras.

A subset of such change detection systems aims to detect abandoned objects, with wide security-related applications in areas such as industrial and airport venues [11]. As they constitute very active research areas, these problems have large image and video datasets at their disposal, encompassing diverse scenarios and objectives. In this context, many solutions have been proposed in the past few years using deep neural networks (DNNs) [12]. These models are pretty data-hungry and take advantage of the abundance of available data to propose solutions with excellent capability to generalize from training to unseen (test) datasets.

However, significantly fewer publications and associated datasets regarding change detection in surveillance videos acquired with moving cameras exist. Most of the available research has been concentrated on monitoring the activity of people and vehicles using pan-tilt-zoom (PTZ) cameras [13]. Distinct approaches for general moving cameras, such as the one investigated in this work, rely on comparing a reference video, without any abnormal activity, and a target video, possibly containing anomalies. The remainder of this section reviews such methods.

Kong et al. [14] describe the approach entitled Detecting Abandoned Objects with a Moving Camera (DAOMC), which uses a modified version of the random sample consensus (RANSAC) [15] algorithm to geometrically align the reference and target videos that were previously synchronized based on GPS data. The geometrically aligned frames are then compared to find suspicious anomalous regions. Finally, the local similarity is analysed to remove false alarms in flat areas, and temporal filtering is used to confirm the presence of anomalies.

Carvalho et al. [16] introduce the Anomaly Detection with a Moving Camera using Multiscale Video Analysis (ADMULT), which is a multi-object detection method for cluttered scenes using a multi-resolution approach. The algorithm also employs normalized cross-correlation and voting steps to avoid false detection and imposes real-time constraints to allow practical use and industrial deployment.

Another method, referred to as the moving-camera background subtraction (MCBS) [17], detects objects in videos

acquired by a camera facing the front of a moving train. The method used background subtraction to detect objects from never-before-seen classes.

The accelerated version of the moving-camera robust subspace recovery (msRoSuRe-A) [18] algorithm uses a sparse-plus-low-rank matrix factorization to isolate video anomalies from a moving background modelled by the subspace associated with the matrix factorization. This method intrinsically performs a reference-target video alignment and uses their cross-correlation function to model the target frames based on the subspace estimated from the corresponding reference frames.

The algorithm entitled Moving-Camera Domain-Transformable Sparse Representations (mcDTSR) [19] improves upon the previous approach by compensating for errors in the video background modelling stage. This is performed by incorporating geometric transformations in the sparse representation process to mitigate the effect of possible spurious camera movements, such as trepidation.

All the above methods train data-driven solutions by relying on traditional (hand-designed) heuristics to overcome the reduced data availability. Two recent approaches using deep convolutional networks deal with the lack of enough training data by relying on pre-trained networks to perform feature extraction for subsequent change-detection classifiers.

The method presented in [20] uses twin convolutional features and learnable morphological operations (TCF-LMO) to perform the change-detection task. This approach reduces the number of features by performing a weighted subtraction of the reference and target tensors, thus losing part of the information obtained in the two Siamese outputs.

For local change detection, where small-scale variations are analyzed in a much wider scene, recent approaches often rely on high-dimensional feature representations to capture subtle changes. Several deep learning architectures have been proposed for generating these representations: Modified ResNet-152 with hybrid pyramidal pooling [21], ResNet-101-based architectures with transfer learning [22] and [23], and ResNet-50 combined with global average pooling [24] have been employed to extract discriminative features. Similarly, improved VGG-19 networks [25] were proposed to enhance feature pooling for precise localization, while multi-scale frameworks, such as [26] for contrast-preserving architectures and end-to-end networks [27], leverage both high- and low-level features for robust detection. Beyond deep learning, kernel-induced possibilistic fuzzy models [28] have also been explored for projecting data into higher-dimensional spaces to improve change detection.

The approach of extracting features from the frames using pre-trained CNNs is in line with other recent methods that rely on CNN-based feature extraction and Random Forest classification, such as [29]. In [30], a CNN helps to expand the number of features to be fed to a tree-based classifier, while in [31], the features extracted by the CNN help to build the Random Forest classifier in a few-shot learning

scenario. Alternatively, other works have relied on other types of decision trees, such as the LightGBM [3] for the classification stage.

A common step to all the previous methods is a reference-target frame-level synchronization. In a few cases, like [18] and [19], such synchronization is intrinsic to the change detection method. However, a more usual approach is to first align both reference and target videos before performing the detection. In addition to the time-alignment techniques used in the previously mentioned methods, other strategies include methods based on optical flow, like RANSAC-Flow [32] and GloFlow [33], based on learning-based temporal-alignment networks employing efficient pattern matching [34], based on feature-level alignment [35] or based on CNNs with deformable convolutions feature-level alignment [36].

The following section presents the proposed approach to the problem of change detection in moving camera videos in a cluttered industrial environment. Although relying on the power of deep learning algorithms, it circumvents the lack of training data by working just with sample-level features. Instead of detecting changes on the bounding box level, it detects changes on the level of pixel-level silhouettes.

III. THE PBCD-MC ALGORITHM

The block diagram of the PBCD-MC algorithm is depicted in Fig. 1. It is composed of five modules: the time-alignment module, which provides for each target frame a corresponding warped reference frame; the feature-extraction module, where deep characteristics of the registered input videos are extracted; the data-sampling module, designed to reduce the data size to speed up the training process; the classification module, which categorizes the samples as anomalous or non-anomalous; and the post-processing module, which is used to exploit spatial and temporal redundancy information to improve detection performance. All these modules are detailed further below.

When compared to the TCF-LMO algorithm described in [20], the newly proposed PBCD-MC algorithm incorporates modifications in all of its modules shown in Fig. 1, and, as a consequence, presents all of the following advantages: (i) The PBCD-MC is tailor-suited for pixel-level change detection (see Section III-A), thus being able to detect the silhouette of the video modifications, as opposed to the TCF-LMO that was optimized to detect only their respective bounding boxes. Due to its pixel-wise classification approach, the PBCD-MC method can detect multiple simultaneous anomalies in a single frame, as it evaluates each pixel independently against the reference video, regardless of the number or spatial distribution of abandoned objects; (ii) A new alignment module is deployed by the PBCD-MC algorithm incorporating an elastic transformation onto the optical flow estimation, as detailed in Section III-B; (iii) The PBCD-MC algorithm simplifies the change-detection problem by categorizing each image pixel instead of the entire video frame. This reduces the problem dimension by several orders of magnitude and allows for network simplification,

as discussed in Section III-D; (iv) Due to the reduction in the problem dimension provided by the categorization of each pixel instead of the frame, the PBCD-MC can successfully combine the reference and target deep features by juxtaposing them into a single tensor (see Section III-C), as opposed to the TCF-LMO, which performs a weighted subtraction of these features and, as a consequence, loses part of their original combined information; (v) The PBCD-MC uses a gradient-boosting LightGBM classifier, which presents a faster training procedure than the random forest employed by the TCF-LMO (see Section III-E); (vi) The post-processing module, as detailed in Section III-F, optimizes the PBCD-MC performance for the silhouette version of the detection problem, possibly allowing one to identify the cause of the change, which is often hindered in the bounding-box case.

A. VIDEO DATASET

The Video Database of Abandoned Objects (VDAO) [37], [38] was designed to emulate abnormal occurrences in a real industrial environment. It illustrates an oil and gas industrial plant environment containing multiple metallic tubes, pipes, cables, valves, and other pieces of machinery. Such a complex and crowded environment provides several challenges for the change-detection problem, namely occlusion, reflections, shadows, and parallax.

This work used a branch of VDAO containing additional ground-truth annotations, called VDAO-AS [39], where the AS stands for *annotated silhouette*. This upgraded dataset is composed of three types of videos: The *reference videos* represent the regular operation of the industrial plant, serving as a comparison basis for all other videos to be analyzed; *target videos* simulate an abnormal behaviour by including one or more abandoned objects that do not appear in the reference videos; the *silhouette videos* were introduced in the VDAO-AS and contain silhouette-level ground-truth annotations associated with each target video. While bounding boxes are effective for representing the general location of anomalous objects, they have limitations in scenarios where the shape of the object is oblong or diagonally oriented. In such cases a significant portion of the bounding box may contain background information, which, in a pixel classification task, can hinder the overall classification performance.

In contrast, using silhouettes allows for a more precise representation of the shape of the anomalous object, excluding unwanted pixels from the training process. From the detected shape, if needed, one can generate a bounding box, but from a bounding box one cannot infer the anomalous object shape.

The PBCD-MC framework was designed for a surveillance scenario where the camera moves through the environment in a predetermined path (defined by the rail where the camera is installed). The nature of the VDAO-AS dataset led to the development of some procedures such as the temporal and the spatial alignment inside the Alignment Module, in order

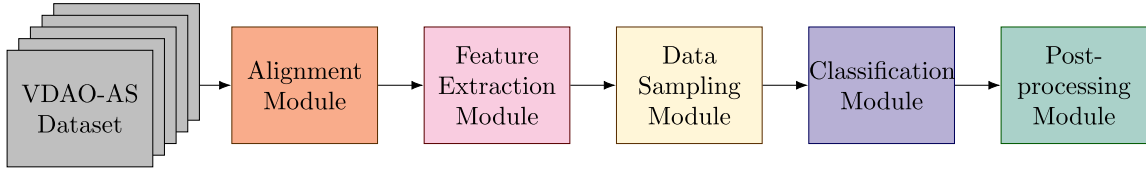


FIGURE 1. Block diagram of the PBCD-MC algorithm. The alignment module registers reference and target frames, both spatially and temporally; the feature-extraction module extracts deep characteristics of the registered input videos; the data-sampling module selects a balanced subset of samples to use during training, optimizing computational resources; the classification module uses machine-learning algorithms to distinguish between normal and anomalous patterns; the post-processing module induces temporal and spatial consistency in the obtained results, improving system accuracy.

to compare the target video frames with previously recorded frames from reference videos.

Other change detection datasets such as UCSD [40] and *changedetection.net* [41], while valuable for other types of change detection tasks, do not exhibit the same motion characteristics as the VDAO-AS dataset. Therefore, a direct comparison with these datasets would not be neither fair or meaningful, as they do not align with the problem the proposed PBCD-MC algorithm is designed to solve. To the best of the author's knowledge, VDAO-AS is the only publicly available dataset with the desired characteristics.

To generate such annotation, a semi-automated method [42] applied a classifier to each reference-target pair of videos to identify pixels that were certainly anomalous or certainly not anomalous, thus determining the anomaly and background silhouettes, respectively. The dubious pixels in the thin frontier between both silhouettes, whose labelling would require a costly fine-grained manual annotation process, were instead labelled as belonging to an uncertainty zone to be disregarded during training and testing procedures. This is illustrated in Fig. 2. It is important to note that using the uncertainty zone speeds up the annotation process without compromising the performance of the trained classifiers since only anomalous pixels are labelled as anomalous, and only background pixels are labelled as background. Only 7.3% of the VDAO-AS pixels were labelled as belonging to the uncertainty zone.

The complete VDAO-AS consists of 59 reference-target-silhouette video trios. Each video was recorded in a cluttered industrial environment using a 720×1280 resolution camera moving over rails in a linear back-and-forth movement. The videos were recorded at different hours of the day, under different lighting conditions, and the anomalous objects were positioned in different scene planes, thus, in many cases, being subjected to partial occlusions. Frames containing the anomalous objects in the VDAO-AS target videos are shown in Fig. 3.

B. ALIGNMENT MODULE

The alignment module is responsible for finding the reference frame R_j that is best temporally and geometrically aligned to each target frame T_i . To estimate the best alignment, consider the reference and target videos as two frame sequences of $H \times W$ pixel resolution. The search algorithm can then be

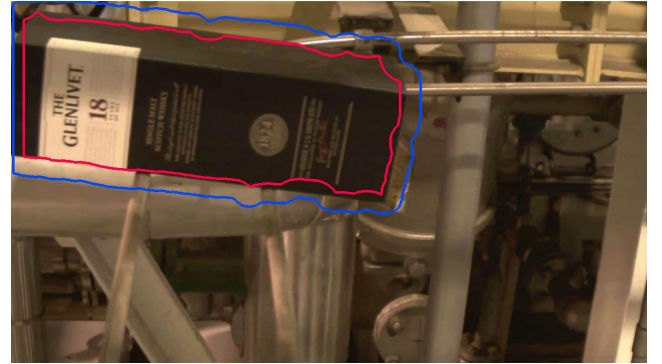


FIGURE 2. Example of anomaly annotation in the VDAO-AS. The red contour represents the silhouette of the anomalous object, whereas the exterior of the blue contour delimits the background (non-anomalous) region. The region between the red and blue contours contains pixels with uncertain classification, which are disregarded during the training and testing of the proposed architecture.

written as

$$j = \underset{k}{\operatorname{argmin}} \sum_{x=1}^W \sum_{y=1}^H [T_i(x, y) - \mathcal{W}_{k,i} \{R_k\}(x, y)]^2, \quad (1)$$

with the current search index k restricted to an interval of length $\Delta = 5$ around the previous index k_p . In this process, $\mathcal{W}_{k,i}$ denotes a geometric warping transformation that best maps the reference frame R_k to the target frame T_i , which is parameterized according to a pre-defined family of transformations.

In this equation, the temporal or inter-frame alignment refers to the search for the point in the camera trajectory which captured a target frame T_i . This search is made by looking at the frames recorded in the reference videos and finding the best match R_j . During this frame search, another alignment, called intra-frame or geometric alignment, tries to compensate for some camera misalignments between T_i and R_j through $\mathcal{W}_{k,i}$.

As the industrial plant contains objects at different depths, considering $\mathcal{W}_{k,i}$ as an elastic deformation allows for simultaneous correction of multiple parallaxes. In this work, an elastic transformation based on the optical flow estimation [43] between R_k and T_i is proposed.

For this matter, the following transformation is considered:

$$\mathcal{W}_{k,i} \{R_k\}(x, y) = R_k(x' + u(x', y'), y' + v(x', y')), \quad (2)$$

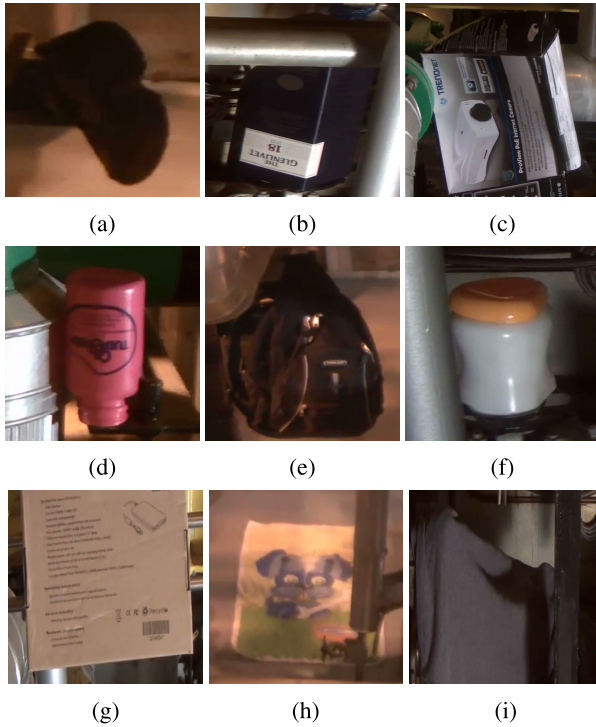


FIGURE 3. Anomalous objects in the target videos of VDAO-AS: (a) shoe (A); (b) dark blue box (B); (c) camera box (C); (d) pink bottle (D); (e) black backpack (E); (f) white jar (F); (g) brown box (G); (h) towel (H); (i) black coat (I).

where (x', y') and (x, y) represent the image coordinates before and after the elastic deformation, respectively, as determined by the vector field defined by the functions $u(\cdot, \cdot)$ and $v(\cdot, \cdot)$ belonging to the parameterized pre-defined family of transformations.

C. FEATURE-EXTRACTION MODULE

Considering a reference image $\hat{R}_j$ already aligned to a target frame T_i , the feature-extraction module shall obtain enough information to distinguish between anomalous and non-anomalous regions. We then propose a feature extraction based on siamese networks [44] composed of two ResNet-50 pre-trained on ImageNet [45], as depicted in Fig. 4. The network is truncated on the third residual block (output of the conv2 layer), which has previously provided the best anomaly-detection performance on the original VDAO dataset [46].

The deep-feature tensors generated in this module output can thus be expressed as

$$F_{R_j} = \text{CNN}(\hat{R}_j) \text{ and } F_{T_i} = \text{CNN}(T_i). \quad (3)$$

Note that at its output layer, the truncated ResNet-50 employed in this work reduces the frame resolution 8-fold and generates 256 features per pixel, such that $F_{R_j}, F_{T_i} \in \mathbb{R}^{90 \times 160 \times 256}$. A tensor F_{R_j, T_i} , created by concatenation in the features dimension of F_{R_j} and F_{T_i} , contains information of

both target and reference frames and totals 90×160 samples with 512 features, as depicted in Fig. 4.

D. DATA-REDUCTION MODULE

The algorithm should produce a binary pixel-level map from the feature tensors indicating whether there is a change or not in the target videos. However, since the feature-extraction module multiplies the input dimension by 64 (spatial resolution drops by a factor of 8 while the number of features increases by 512), the use of the complete feature tensor would require a classifier with a large number of trainable parameters, preventing it from being used in data-constrained scenarios. Bruno et al. [46] addresses this issue by directly subtracting reference and target features to reduce the feature dimension by half, while Padilla et al. [20] proposes a lightweight architecture with few trainable parameters to combine the reference and target features, but still reducing the feature dimension by a half.

Differently from these two approaches, this work proposes a change of paradigm: instead of working with feature vectors at the image level, that is, belonging to $\mathbb{R}^{90 \times 160 \times 512}$, one works with vectors at the sample level, that is, belonging to $\mathbb{R}^{512}$.

In other words, for a $90 \times 160 \times 512$ feature tensor, each sample is treated independently of the others, resulting in the classifier being trained for each frame of the reference and target videos on 90×160 feature vectors of dimensions 1×512 . Such a procedure drastically increases the size of the training dataset but also introduces too much redundant information, making the training of the classification module impractical. Since the video frames are naturally redundant, they can be subsampled and yet retain most information that is relevant for their classification. Therefore, in this work, one takes only $N = 144$ random samples of each frame (1% of its original size). Such a subset of samples has been experimentally shown to contain sufficient information to discriminate the anomalies, while being sufficiently small to allow a computationally feasible classifier training. In this sampling procedure, $\frac{N}{2}$ random samples are selected from each of the background and anomalous regions, avoiding samples from the uncertainty region, as shown in Fig. 5. The 1×512 feature vector is used to train the classifier for all selected samples, as depicted in Fig. 6.

It should be noted, however, that this sampling procedure is applied only during training. All 90×160 samples from the image tensor are fed to the classification module during inference or testing.

E. CLASSIFICATION MODULE

The classification module receives the feature vectors and decides if the corresponding samples refer to a change or not. Tree-based classifiers are among the most widely used models due to their robustness and ability to select and highlight important features [29], [30], [31], [47]. For this reason, this work considers the gradient-boosting LightGBM

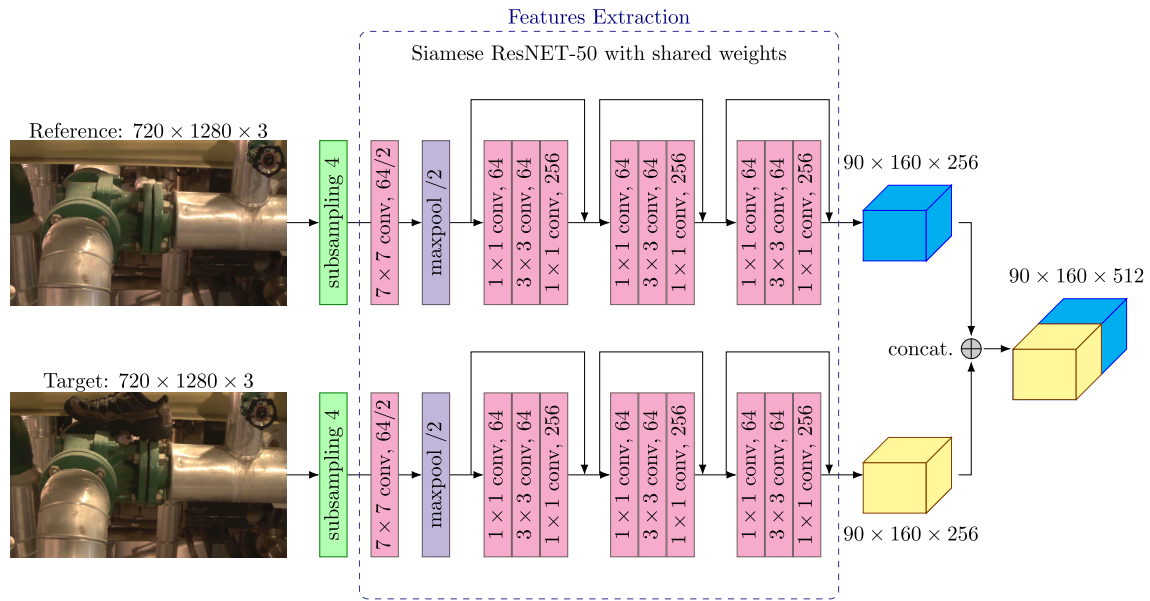


FIGURE 4. Block diagram of feature-extraction module, showing the two siamese ResNet-50 applied to the pre-aligned reference and target frames. The extracted features are concatenated, generating a $90 \times 160 \times 512$ tensor.

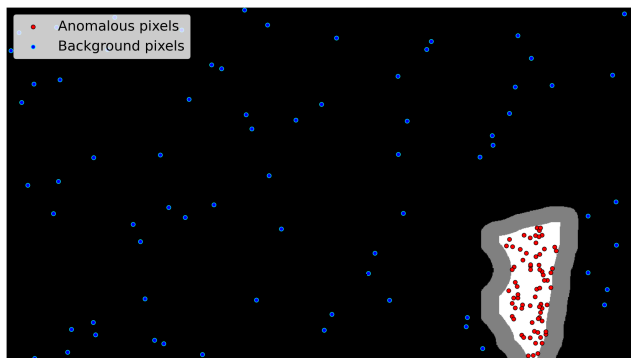


FIGURE 5. Example of pixel sampling applied during classifier training. The uncertainty (grey) regions are ignored, and $\frac{N}{2} = 72$ pixels are randomly selected from each of the anomalous (red dots) and background (blue dots) regions.

framework [3], [48], which is a gradient boosting algorithm based on decision trees, designed to be a lightweight and efficient solution. It achieves this by combining two key strategies: leaf-wise tree growth and histogram-based splitting. Leaf-wise tree growth consists of, instead of growing trees level-by-level, selectively splitting the leaf that contributes the most to reducing the loss gradient. This approach prevents the trees from becoming excessively large while improving model accuracy. Histogram-based splitting means further optimizing performance by converting continuous feature values into discrete bins (defined by one histogram per feature). This reduces the computational load, especially for large datasets, by limiting the number of split points that need to be evaluated. Together, these strategies make LightGBM highly efficient and scalable for training on large datasets.

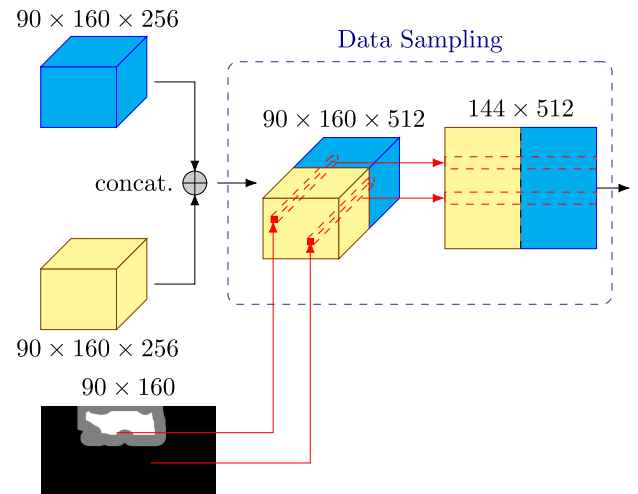


FIGURE 6. Schematic of the data reduction module indicating how during training, the original $90 \times 160 \times 512$ feature tensor is reduced to $N = 144$ feature vectors of dimensions 1×512 each.

The LightGBM hyperparameters were optimized using a Bayesian search [49], using the *scikit-optimize* [50] Python library, on the ranges shown in Table 1.

F. POST-PROCESSING MODULE

In the final step of the proposed architecture, a post-processing module enforces spatio-temporal consistency of the classified pixels. Such consistency is expected in a real application, given that the anomalies, in this case, abandoned

TABLE 1. List of the trainable hyperparameters from the LightGBM classifier.

Hyperparameter	Min	Max	scale	Step
Number of leaves (N_l)	1	256	uniform	1
Minimum samples per child (M_s)	1	200	uniform	1
Classification threshold (τ)	0	1	uniform	0.01

objects, do not show chaotic behaviour throughout the recorded videos. This is so because they tend to concentrate in a few specific regions within a frame, and they do not undergo abrupt displacements in consecutive frames.

The spatial or intra-frame consistency is improved through morphological operations such as opening and closing. The opening operation mainly removes small and scattered false-positive regions that may appear due to misalignment, lighting differences, reflective surfaces, or occlusions. After the opening operation, the closing operation is used to remove small false-negative regions that may occur, for instance, on objects that tend to blend with the background, as illustrated in Fig. 7.

For the temporal or inter-frame consistency, a $w \times w$ voting window along d consecutive frames is employed on the classified pixels. If there is at least a fraction v of the $w \times w \times d$ parallelepiped filled, its central pixel is considered to be anomalous.

The post-processing hyperparameters, including the kernel sizes o and c for the opening and closing operations, respectively, and the width w , depth d , and threshold v of the voting window, were all tuned through a Bayesian search along the ranges specified in Table 2.

TABLE 2. List of trainable hyperparameters from the post-processing module.

Hyperparameter	Min	Max	scale	Step
Classification threshold (t)	0	1	uniform	0.01
Opening size (o)	1	70	uniform	1
Closing size (c)	1	70	uniform	1
Voting window width (w)	1	70	uniform	1
Voting window depth (d)	1	70	uniform	1
Voting window threshold (v)	0.01	1	uniform	0.01

IV. ANALYSIS OF THE PROPOSED SYSTEM

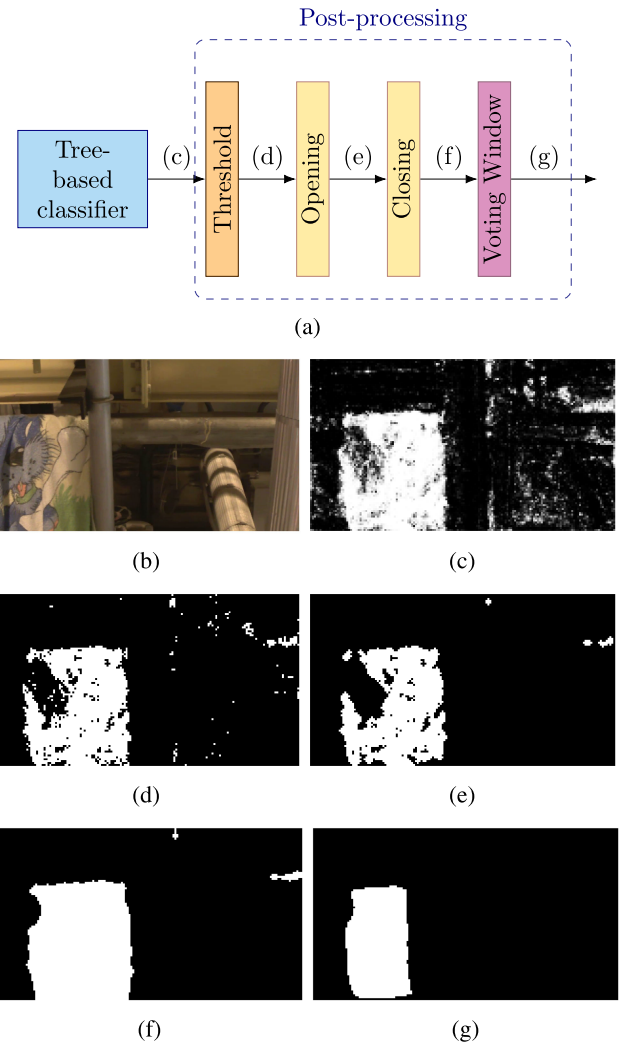
A. EXPERIMENTAL CONDITIONS

Some strategies were adopted to reduce the computational complexity of the proposed framework to facilitate its training. These strategies are listed below.

1) TRAINING AND TEST DATASETS

The first strategy was the choice of the test and training datasets.

For the test set, an extension of the VDAO-200 dataset [51] that adds silhouette annotations is considered. This dataset contains a representative subset of videos with balanced

**FIGURE 7.** Illustration of the classification and post-processing procedures of the PBCD-MC algorithm: (a) schematic of the classification and post-processing modules; (b) target input frame; (c) tree-based classifier output, where for each pixel an anomaly score between 0 and 1 is produced; (d) binarized classifier output; (e) opening operator output; (f) closing operator output; (g) output of the voting window operation.

numbers of frames containing and not containing anomalies. For each target video, a 200-frame excerpt in different positions and illumination conditions, including at least one of the objects shown in Fig. 3, is extracted to compose this extension of the VDAO-200 dataset. This test set is referred to as VDAO-AS-test.

In addition, to reduce the amount of training samples, a novel VDAO subset, comprising 200-frame excerpts on distinct positions from the test set, is employed. This test set is referred to as VDAO-AS-training. Due to the proposed pixel-based training, only frames that contain anomalies are selected. These frames are then spatially sampled to produce $\frac{N}{2} = 72$ anomalous and $\frac{N}{2} = 72$ non-anomalous pixels per frame (see Section III-D). The employed training (VDAO-AS-training) and test sets (VDAO-AS-test), along with the

reference-target aligned videos and the proposed model, are available at https://github.com/lgtavares/VDAO_Pixel.

2) CROSS-VALIDATION PROCEDURE

A cross-validation procedure is performed for hyperparameter tuning and to evaluate the system's generalization capabilities. This procedure was implemented by dividing the training and validation dataset into $K = 9$ folds, each one corresponding to one of the 9 object types in the VDAO single-object target videos (see Fig. 3). Likewise, the VDAO-AS-test video set is also segmented according to the object it contains. For each fold k , there is an associated subset t_k of the VDAO-AS-test composed only of videos containing the corresponding object class. To avoid data contamination during testing, when subset j of the VDAO-AS-test is used for testing, training and validation are performed with the VDAO-AS-training videos from the eight remaining folds $k \neq j$. This separation guarantees that, for each fold, eight objects are seen by the classifier during training and validation, while one object remains unseen and is then used for testing.

The training and validation data are themselves also split into a nested K -fold configuration for tuning the system hyperparameters. Due to computational and time limitations, a separation corresponding to five objects for training and three for validation was used, which allows a total of $\binom{5}{3} = 56$ possible combinations. However, to reduce the training time, only eight combinations out of the 56 possible were chosen. This choice was made with the care that each fold corresponding to a given object was used five times in training and three times in validation. This nested cross-validation procedure is illustrated in Fig. 8.

B. EVALUATION STRATEGIES

To train the proposed modules and evaluate their performances, we follow three paradigms that determine what is considered a successful detection: a frame-level evaluation, an object-level evaluation, and a novel pixel-level evaluation.

1) FRAME-LEVEL EVALUATION

This strategy extrapolates anomaly detection masks to classify each frame as a whole. A frame is treated as a ground-truth positive if it contains at least one silhouette annotation for a spurious object. An input frame is associated with a true-positive (TP) detection if the estimated mask is present on a ground-truth positive frame and intersects any pixel outside the uncertainty region silhouette. Any detection in a non-anomalous frame is treated as a false positive (FP), whereas the absence of detection in such frames counts as a true negative (TN). The absence of detection (or the presence of estimation masks that only intercept the uncertainty region silhouette) in an anomalous frame is counted as a false negative (FN). Therefore, as outlined above, frame-level TP, FP, TN, and FN can be computed by comparing the change estimates with the ground-truth annotations in each frame.

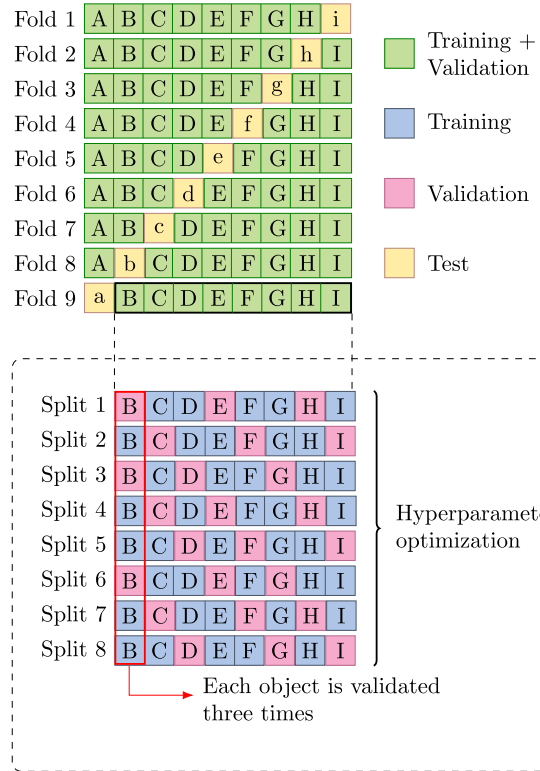


FIGURE 8. Illustration of the nested cross-validation procedure. When the videos of the VDAO-AS-test corresponding to a given object are used for system testing, the folds of the VDAO-AS-training corresponding to the remaining eight objects are used either in system training (5 times) or validation (3 times). The objects are represented by the letters using the code from Fig. 3 caption.

2) OBJECT-LEVEL EVALUATION

this strategy evaluates if the anomaly estimation masks intersect with the annotated silhouettes for all individual spurious objects. The object-level TP, FP, TN, and FN metrics are determined analogously to their frame-level counterparts by comparing the change estimates with the ground-truth annotation for each object.

If a target frame includes n_o spurious objects, it counts only once for the frame-level metrics and n_o times for the object-level strategy.

3) PIXEL-LEVEL EVALUATION

This strategy performs a fine-grained evaluation by examining the prediction accuracy for each image pixel. In it, for each frame, the samples in the uncertainty zone are ignored; that is, the TP, FP, TN, and FN are computed considering only the samples not in the uncertainty zone. The TP is computed as the percentage of anomalous pixel samples correctly classified as anomalous, the FP as the percentage of non-anomalous pixels incorrectly classified as anomalous, the TN as the percentage of non-anomalous pixels correctly classified as non-anomalous, and the FN as the percentage of anomalous pixels incorrectly classified as non-anomalous.

While the frame-level and object-level evaluations are more sensitive to errors, the pixel-level evaluation provides a fairer and more granular analysis. As an example, let's consider a single frame, where an anomalous object is present and correctly detected ($DIS = 0$ and $MCC = 1$). In this same example, let us consider now that a single false positive pixel is present in the entire image. In this case, using the object-level evaluation, $TP = 1$, $FP = 1$, $FN = 0$, $TN = 1$, which results in a drop of the object-level metrics to $DIS = 0.5$ and $MCC = 0.5$, which is far from perfect detection. This example illustrates how frame-level and object-level metrics can be disproportionately affected by minor errors, even when the detection is largely accurate. By focusing on pixel-level evaluation, we aim to provide a more robust and meaningful assessment of the proposed method's performance.

C. EVALUATION METRICS

In this work, the Matthew correlation coefficient (MCC) [52], [53], [54], [55] is used both as a loss function and as a metric to evaluate the detection performance. It is given by

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \quad (4)$$

with TP, FP, TN, and FN defined according to the evaluation strategy. For the loss function, such values are used on a pixel-level basis. In our two-class case, the MCC works as a cosine distance between the predicted data and the ground truth. For the degenerate cases, when the denominator in Eq. (4) is 0, the MCC is also defined as 0. The MCC is in the interval $[-1, 1]$, and the larger it is, the better the classifier is. One advantage of the MCC is that in the case of a classifier that just randomly outputs TP, FP, TN, and FN detections with the probabilities of the ground truth data, the MCC is zero.

To enable the comparison with previously published state-of-the-art methods, the system performance is also evaluated for each strategy (frame, object, or pixel) by determining the best compromise between the TP and FP rates (TPR and FPR, respectively) using the DIS metric,

$$DIS = \sqrt{(1 - TPR)^2 + FPR^2}, \quad (5)$$

which represents the minimum distance of the obtained (TPR, FPR) point to the one corresponding to the ideal performance on the $TPR \times FPR$ plane, $(TPR, FPR) = (1, 0)$.

The DIS and MCC values are integrated across all individual test videos to produce three distinct metrics: the average value considers the average of the DIS/MCC values obtained for each video (equivalently for the median value), whereas the overall value acknowledges that, despite the test set having a balanced amount of anomalous and non-anomalous frames, each individual video does not; it is then equivalent to concatenating all 59 test videos into a single video to obtain the total TPR and FPR used in Eqs. (4) and (5).

D. HYPERPARAMETER TUNING STRATEGY

Due to limitations in memory and processing time, the hyperparameters concerning both the classification and the

post-processing stages were tuned through two separate Bayesian searches.

The first one attempts to maximize system performance by adjusting, in the LightGBM classifier, the number of leaves, the minimum number of samples per child, and the classification threshold, with search limits as shown in Table 1. The second search tunes the post-processing parameters with the search limits as given in Table 2 and also re-optimizes the classification threshold. Each search epoch attempts to maximize the pixel-level MCC averaged across all eight splits in each fold, as depicted in Fig. 8.

Table 3 shows the best combination of post-processing parameters resulting from the performed Bayesian search that suits all folds: a *classification threshold* of ≈ 0.65 that reduces the FP and increases the FN detection regions, a small *opening size* $o \approx 4$ and a large *closing size* $c \approx 30$, with the voting window characterized by $(w, d, v) \approx (28, 9, 0.45)$ further reducing the false-detection rates.

TABLE 3. List of the best set of hyperparameters found for each fold in the proposed configuration. The hyperparameters and their search range are defined in Tables 1 and 2.

Fold	Classification			Post-Processing				
	N_l	M_s	τ	Morphology		Voting window		
				o	c	w	d	v
1	153	200	0.58	4	27	9	32	0.48
2	256	200	0.60	4	27	11	22	0.46
3	186	200	0.95	2	31	1	43	0.46
4	191	174	0.86	2	9	10	12	0.44
5	182	200	0.65	4	29	11	22	0.41
6	166	174	0.63	4	27	9	28	0.40
7	155	197	0.44	5	26	7	28	0.45
8	181	200	0.91	1	12	15	23	0.39
9	92	144	0.83	3	19	8	29	0.38

V. PERFORMANCE EVALUATION

This section assesses the proposed PBCD-MC algorithm's performance against the previously published algorithms on the VDAO dataset [51]. These algorithms have been briefly described in Section II. It is important to note that, for a fair comparison, the classification results of such methods were obtained by running their available software for the new VDAO-AS-test set, which considers a revised reference-target alignment and a new silhouette-level annotation. All metrics are computed as described in Section IV-B, using the novel silhouette-level annotation, that is, disregarding all pixels within the uncertain region.

Table 4 compares the TPR and FPR individual metrics for the discussed methods on the VDAO-AS-test set. Since those values present a trade-off behaviour, they are integrated into the MCC (see Eq. ((4))) and DIS (see Eq. ((5))) metrics to better evaluate the overall algorithm's classification performance and determine how good such trade-off is. The MCC and DIS results are shown respectively in Tables 5 and 6. From the TPR and FPR individual metrics, one can see that the proposed algorithm outperforms all the previously

published works on the VDAO dataset in the pixel-level evaluation scheme for which it was designed, while on the frame- and object-level scheme, other methods achieved higher scores, although not simultaneously on both TPR and FPR metrics. By aggregating such values on the DIS and MCC metrics, one can see that the proposed algorithm offers the best trade-off: for the pixel-level, frame-level, and object-level evaluation schemes, the PBCD-MC algorithm significantly outperforms all other algorithms in terms of both MCC and DIS.

TABLE 4. Performance comparison, based on the TPR and FPR metrics, of proposed PBCD-MC algorithm against other works on the VDAO-AS-test set for the frame-level, object-level, and pixel-level evaluation schemes. Bold numbers indicate the best performance according to each evaluation scheme.

	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
Frame-Level Evaluation						
TPR	0.513	0.750	0.994	1.000	0.974	0.863
FPR	0.149	0.329	0.993	1.000	0.744	0.225
Object-Level Evaluation						
TPR	0.503	0.705	0.217	0.832	0.958	0.845
FPR	0.206	0.657	0.997	1.000	0.906	0.341
Pixel-Level Evaluation						
TPR	0.675	0.651	0.025	0.580	0.809	0.907
FPR	0.005	0.005	0.016	0.423	0.020	0.004

TABLE 5. Performance comparison, based on the MCC metric, of the proposed PBCD-MC algorithm against other works on the VDAO-AS-test set for the frame-level, object-level, and pixel-level evaluation schemes. Bold numbers indicate the best performance according to each evaluation scheme.

	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
Frame-Level Evaluation						
Average	0.284	0.353	0.003	0.000	0.223	0.546
Median	0.269	0.511	0.000	0.000	0.000	0.762
Overall	0.327	0.388	0.004	0.000	0.360	0.638
Object-Level Evaluation						
Average	0.259	0.086	-0.849	-0.249	0.126	0.485
Median	0.247	0.038	-0.950	-0.131	0.000	0.737
Overall	0.271	0.050	-0.826	-0.323	0.103	0.517
Pixel-Level Evaluation						
Average	0.423	0.538	0.003	0.016	0.570	0.724
Median	0.454	0.622	-0.006	-0.010	0.607	0.855
Overall	0.699	0.688	0.011	0.047	0.612	0.877

To compare model complexity, we used average prediction time as the primary metric. This choice was motivated by the diverse image processing algorithms employed by the compared methods: ADMULT [16] relies on cross-correlation between images, mcDTSR [19] and MCBS [17] are based on sparse representations, and DAOMC [14] utilizes homographies. Only TCF-LMO [20] and the proposed PBCD-MC method use Convolutional Neural Networks. It is important to note that, for a fair comparison, the classification results

TABLE 6. Performance comparison, based on the DIS metric, of the proposed PBCD-MC algorithm against other works on the VDAO-AS-test set for the frame-level, object-level, and pixel-level evaluation schemes. Bold numbers indicate the best performance according to each evaluation scheme.

	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
Frame-Level Evaluation						
Average	0.559	0.434	0.860	0.864	0.672	0.324
Median	0.539	0.295	1.000	1.000	0.872	0.152
Overall	0.509	0.413	0.993	1.000	0.744	0.264
Object-Level Evaluation						
Average	0.570	0.716	1.311	1.047	0.840	0.402
Median	0.539	0.768	1.366	1.000	1.000	0.189
Overall	0.538	0.721	1.268	1.014	0.907	0.374
Pixel-Level Evaluation						
Average	0.586	0.516	0.985	0.738	0.239	0.221
Median	0.500	0.514	0.999	0.769	0.144	0.062
Overall	0.325	0.348	0.975	0.596	0.192	0.092

of such methods were obtained by running their available software for the new VDAO-AS-test set, which considers a revised reference-target alignment and a new silhouette-level annotation. These results differ from previous results of these methods, that were obtained on the VDAO dataset, in which the silhouette annotations are not available. Considering the diverse nature of the approaches, prediction time offers a consistent and fair basis for evaluating computational efficiency across methodologies. All runs were done under the same computational power (an Intel Core i7-4790K CPU, 4.00GHz, with 32GB of RAM, and a dedicated NVIDIA GeForce GTX Titan XP with 12GB of VRAM).

A comparison of the average processing time on the VDAO-AS-test dataset is given in Table 7.

TABLE 7. Comparison of the average processing time for the VDAO-AS-test videos (in seconds) of the proposed PBCD-MC algorithm against the ones of other works on the VDAO-AS-test set.

DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
284	114	50543	1050	3.7	34.1

Figs. 11(a) to 11(d) show examples of accurate anomaly masks obtained with the proposed PBCD-MC algorithm. Note that, due to the training using fine-grained pixel-level annotations, the anomaly masks can represent the correct silhouettes of the objects. Figs. 11(e) to 11(h) show examples where the PBCD-MC algorithm provides slightly displaced anomaly masks; yet, these slightly displaced anomaly masks enable the identification of the presence of anomalous objects.

Fig. 9 show a comparative example of the resulting anomaly detection masks between the PBCD-MC against other works.

Based on the visual predictions in Fig. 9, PBCD-MC correctly detects the anomalous bottle with minimal false

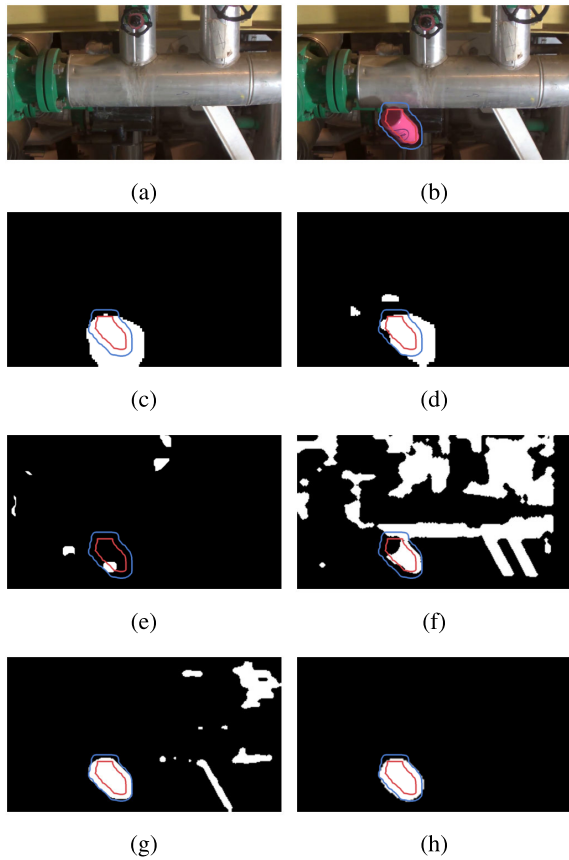


FIGURE 9. Comparison between anomaly detection methods in detecting an anomalous pink bottle. (a) Reference frame; (b) target frame; (c) DAOMC; (d) ADMULT; (e) MCBS; (f) mcDTSR; (g) TCF-LMO; (h) PBCD-MC (proposed).

positives. In contrast, DAOMC, ADMULT, and TCF-LMO achieve detection but with significantly higher FP counts, while MCBS and mcDTSR fail to detect the anomaly altogether.

Another example can be seen in Fig. 10. In this example, PBCD-MC correctly identifies the shoe, whereas DAOMC, ADMULT, and TCF-LMO produce both higher true positives and false positives detections. This discrepancy may come from shadows cast by the shoe on the pipes, which introduce inconsistencies between the reference and the target frame.

One of the analyses that can be made to measure the robustness of the model, its generalization to different circumstances, as well as identifying weaknesses, is to submit the model to different scenarios. In this work, this analysis is made by separating the VDAO-AS-test videos into different categories, such as different illumination conditions, object placement, etc. In this work, three different cases were analysed.

- *Illumination differences:* The test videos were separated into the ones recorded under ambient light and those recorded using extra illumination;

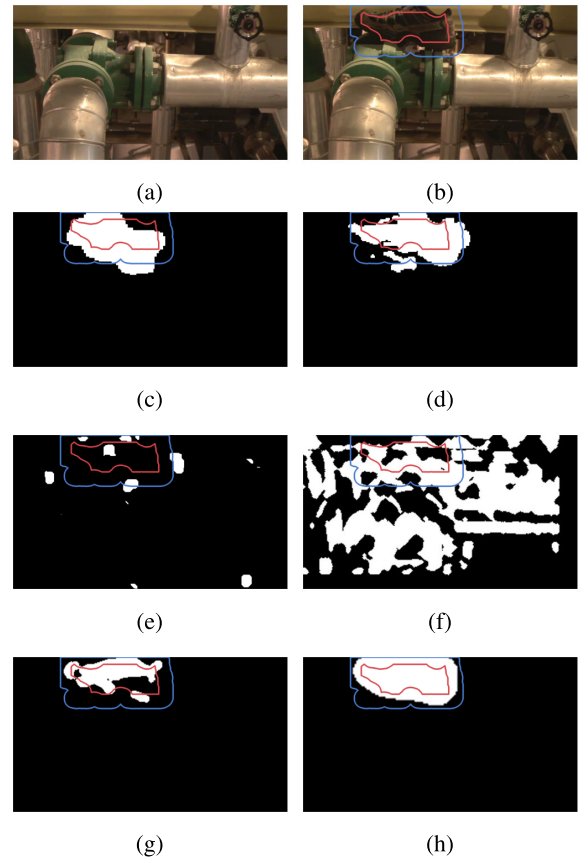


FIGURE 10. Comparison between anomaly detection methods in detecting an anomalous shoe. (a) Reference frame; (b) target frame; (c) DAOMC; (d) ADMULT; (e) MCBS; (f) mcDTSR; (g) TCF-LMO; (h) PBCD-MC (proposed).

- *Object size:* The test videos were visually sorted into three categories: containing small objects, containing objects of average size and containing large objects;
- *Object position:* As the objects on VDAO were placed in different positions on the plant, and each position has its own ambient complexity, the test videos were separated into videos with objects on Position 1 (near a stair), Position 2 (near a green pipe knee) and Position 3 (near an open space behind pipes).

Table 8 refers to the test results discriminating illumination differences, while Table 9 and Table 10 show the results for different object sizes and positions, respectively.

According to Table 8, the usage of extra illumination, in contrast with the ambient light, leads to a poorer performance for all methods. This fact is explained by the additional reflections on metallic surfaces and by more shadows casted on the ambient, leading to more spurious differences, that are not anomalies, between reference and target frames. However, despite the noticeable drop in MCC, the PBCD-MC still holds the highest pixel-level MCC values, compared to other methods.

Comparing the methods when dealing with objects of different sizes, as seen on Table 9, one can notice that, for

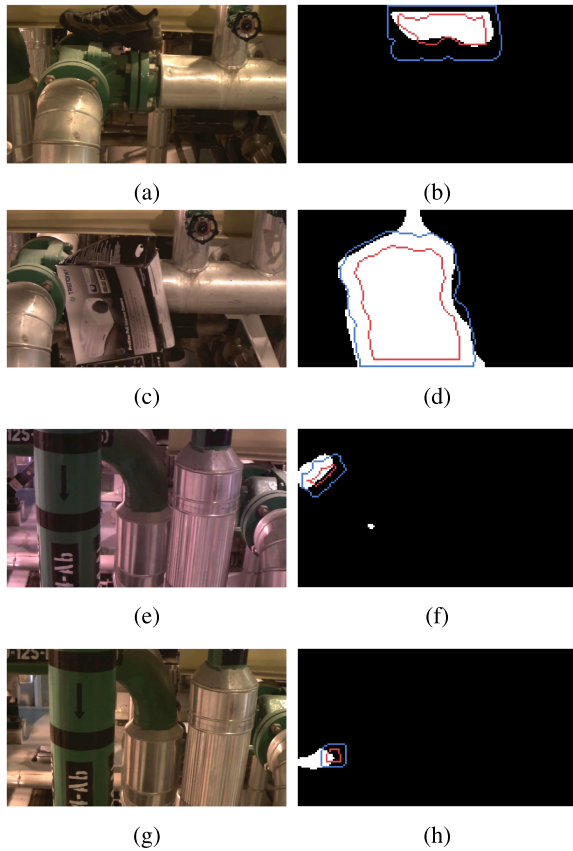


FIGURE 11. Examples of anomaly masks obtained by the proposed PBCD-MC algorithm. Left: target frames; Right: estimated anomaly masks (white) and ground truth annotations for foreground (red) and background (blue).

TABLE 8. Performance comparison, based on the MCC pixel-level metric, of the proposed PBCD-MC algorithm against other works on the VDAO-AS-test set on detecting anomalies under different illumination settings (*Light* column). Bold numbers indicate the best performance according to each evaluation scheme.

Light	MCC	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
Ambient	Average	0.44	0.57	0.01	0.03	0.58	0.74
	Median	0.53	0.68	0.00	0.00	0.61	0.84
	Overall	0.74	0.74	0.01	0.09	0.70	0.91
Extra	Average	0.40	0.51	0.00	0.00	0.56	0.71
	Median	0.43	0.53	-0.01	-0.02	0.60	0.88
	Overall	0.55	0.52	0.00	0.00	0.47	0.78

every compared method, the detection is more difficult for smaller objects, leading to a significant drop in the pixel-level MCC. One can see that, even for smaller objects, PBCD-MC still detects the objects with an overall pixel-level MCC of 0.80.

When discriminating the object position on the scene, one can point out that, for all compared methods, the detection performance is higher when the objects are placed on Positions 1 and 2. This means the detection is easier for these two positions, while Position 3 is the hardest. The detection difficulty for Position 3 can be explained by a

TABLE 9. Performance comparison, based on the MCC pixel-level metric, of the proposed PBCD-MC algorithm against other works on the VDAO-AS-test set on detecting anomalies of different sizes. Bold numbers indicate the best performance according to each evaluation scheme.

Size	MCC	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
Small	Average	0.24	0.52	0.00	0.02	0.51	0.64
	Median	0.00	0.57	-0.01	-0.01	0.51	0.80
	Overall	0.50	0.52	0.01	0.03	0.45	0.80
Medium	Average	0.44	0.50	0.00	-0.01	0.52	0.72
	Median	0.49	0.55	0.00	-0.03	0.48	0.86
	Overall	0.49	0.50	0.01	-0.01	0.42	0.73
Large	Average	0.66	0.64	0.00	0.06	0.75	0.86
	Median	0.74	0.75	-0.01	0.03	0.82	0.93
	Overall	0.79	0.77	0.01	0.10	0.73	0.93

TABLE 10. Performance comparison, based on the MCC pixel-level metric, of the proposed PBCD-MC algorithm against other works on the VDAO-AS-test set on detecting anomalies in different positions of the industrial plant. Position 1 refers to the region near a stair, Position 2 refers to the region near a green pipe knee, and Position 3 refers to an open space behind pipes. Bold numbers indicate the best performance according to each evaluation scheme.

Position	MCC	DAOMC [14]	ADMULT [16]	MCBS [17]	mcDTSR [19]	TCF-LMO [20]	PBCD-MC (proposed)
1	Average	0.44	0.46	0.00	-0.01	0.58	0.70
	Median	0.51	0.44	0.00	-0.03	0.61	0.79
	Overall	0.67	0.66	-0.01	0.04	0.75	0.77
2	Average	0.66	0.64	0.01	0.05	0.71	0.87
	Median	0.71	0.70	-0.01	0.06	0.79	0.94
	Overall	0.73	0.72	0.02	0.07	0.65	0.93
3	Average	0.14	0.49	0.00	0.00	0.40	0.59
	Median	0.00	0.55	-0.01	-0.01	0.39	0.76
	Overall	0.30	0.44	0.00	0.01	0.26	0.75

higher number of metallic pipes on the foreground and more depth in the scene, which increases the parallax in the scene, which leads to the anomalies being more frequently occluded or in the background. This can be confirmed by examining Table 10.

While the compared methods present a significant drop in MCC at pixel-level, the PBCD-MC still maintains an overall $MCC = 0.77$ for Position 1 cases, where one can conclude that PBCD-MC is a more suitable detector for identifying anomalies on background, even in a cluttered industrial scenario.

A. ABLATION STUDIES

To identify the effect of each module in the proposed algorithm as depicted in Fig. 1, this section performs ablation studies. They consist of replacing or removing some of its modules and computing the pixel-level MCC and DIS overall values for the resulting architecture. The modules changed for the ablations are the alignment module, the classification module and the post-processing module, as they contain the most relevant proposed methods in the classification pipeline.

The feature extraction module was not included in the studies as its primary role is to produce 512 features per pixel, which are then passed to the Classification module to weight these features for anomaly detection. The choice of

the feature extractor (a Siamese ResNet-50 architecture) was based on an extensive comparison detailed in [46]. Since the focus of this paper is on leveraging these features to improve anomaly detection in moving-camera scenarios, discussions on the specific backbone architecture and the layer chosen for feature extraction were considered out of scope.

The primary role of the data reduction module is to make the training process computationally viable. While its removal might theoretically lead to a more complete understanding of the characteristics of the proposed method, doing so would be beyond the available computational resources. Therefore, as its role is strictly tied to enabling the training process rather than directly influencing the detection performance, this module was also not included in the ablation studies.

1) ALIGNMENT MODULE

The use of the proposed elastic alignment based on optical flow described in Section III-B is compared to the use of two other alignment methods. The first one is the temporal alignment proposed in [20], which considers a simple identity-warping transformation. Referring to Eq. ((1)), it is such that

$$\mathcal{W}_{k,i}\{R_k\} = R_k. \quad (6)$$

The second alignment method employs a homography transformation $\mathbf{H}$ that attempts to warp the reference frame R_k onto the target frame T_i . Referring to in Eq. ((1)), it is such that

$$\mathcal{W}_{k,i}\{R_k\}(x, y) = R_k(\mathbf{H}\mathbf{x}'). \quad (7)$$

Table 11 shows that the proposed optical flow-based alignment obtains the best results according to the pixel-level MCC and DIS metrics, confirming its effectiveness.

TABLE 11. Ablation study on the alignment module using the VDAO-AS-test set, considering the MCC and DIS provided by the three distinct alignment approaches from Subsection V-A1.

	Temporal	Homography	Optical Flow (proposed)
MCC	0.833	0.867	0.877
DIS	0.140	0.155	0.092

2) CLASSIFICATION MODULE

The use of the proposed LightGBM classification module described in Section III-E is compared to the use of the Random Forest classifier employed in [46], which was optimized following the same hyperparameter ranges in Table 1. The results shown in Table 12 demonstrate the superior performance of the LightGBM classifier on the proposed architecture, confirming its effectiveness.

3) POST-PROCESSING MODULE

The use of the proposed post-processing module configuration described in Section III-F is compared to the use of the following different post-processing configurations:

TABLE 12. Ablation study on the classification module using the VDAO-AS-test set, considering the MCC and DIS provided by the two distinct classification schemes from Subsection V-A2.

	Random Forest	LightGBM (proposed)
MCC	0.722	0.877
DIS	0.426	0.092

- (I) removing only the opening operation ($o = 1$);
- (II) removing only the closing operation ($c = 1$);
- (III) removing both closing and opening operations ($o = c = 1$);
- (IV) removing only the voting window depth ($d = 1$);
- (V) removing only the voting window width ($w = 1$);
- (VI) setting the threshold on the fraction of pixels in the voting window to $v = 0.5$;
- (VII) removing the voting window ($d = w = 1$);
- (VIII) removing the post-processing module altogether, thus not refining the classification results.

The results shown in Table 13 indicate that the proposed post-processing architecture (with complete morphology operations and voting window) consistently achieves a higher MCC value with only a small penalty on the pixel-level DIS value.

TABLE 13. Ablation study on the post-processing module using the VDAO-AS-test set, considering the MCC and DIS provided by the nine different module configurations from Subsection V-A3.

Configuration	MCC	DIS
(I)	0.560	0.061
(II)	0.824	0.264
(III)	0.851	0.204
(IV)	0.860	0.087
(V)	0.869	0.077
(VI)	0.875	0.086
(VII)	0.848	0.082
(VIII)	0.723	0.239
Proposed	0.877	0.092

VI. CONCLUSION

This paper described a pipeline combining high-level deep features and tree-based algorithms in a change detection paradigm. The proposed system combines multiple contributions, as it introduces a complexity-reducing strategy and a pixel-level training methodology particularly suited for data-restricted scenarios, achieving an overall $MCC = 0.877$ and an overall $DIS = 0.092$ for pixel-level evaluations, outperforming the compared state-of-the-art models. Its alignment module implements a flexible deformation technique, which significantly improves the detection accuracy. For the classification module, a LightGBM architecture was shown to be the most suited for the limited database. An analysis of the training process and a detailed ablation study show how the combination of all parts improves upon the final detection results.

A performance evaluation using the VDAO dataset was made by comparing the previous methods using multiple metrics. Results indicate that the proposed pipeline surpasses the performance of current state-of-the-art models with the second-to-best processing time, which demonstrates the successful combination of silhouette annotation, proper data sampling, pixel-level strategy, and efficient LightGBM classifier in the proposed pipeline.

There are, however, many improvements that could be addressed in future works to enhance PBCD-MC performance and efficiency. One priority is reducing model complexity by optimizing the feature extraction process and exploring simpler, more lightweight classifier models, which would significantly decrease the average processing time. Additionally, the issue of border false-positive detections caused by a large kernel for the closing operation, as observed in Fig. 11(d) and Fig 11(f), can be further addressed. Lastly, expanding the dataset and incorporating diverse scenarios could enhance the model's robustness and generalization capabilities, ensuring better performance across varying conditions.

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REFERENCES

- [1] M. Sabokrou, M. Khaloeei, M. Fathy, and E. Adeli, "Adversarially learned one-class classifier for novelty detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 3379–3388.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778, doi: [10.1109/CVPR.2016.90](#).
- [3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–12.
- [4] M. Yazdi and T. Bouwmans, "New trends on moving object detection in video images captured by a moving camera: A survey," *Comput. Sci. Rev.*, vol. 28, pp. 157–177, May 2018.
- [5] R. J. Radke, S. Andra, O. Al-Kofahi, and B. Roysam, "Image change detection algorithms: A systematic survey," *IEEE Trans. Image Process.*, vol. 14, no. 3, pp. 294–307, Mar. 2005, doi: [10.1109/TIP.2004.838698](#).
- [6] T. Minematsu, H. Uchiyama, A. Shimada, H. Nagahara, and R.-I. Taniguchi, "Evaluation of foreground detection methodology for a moving camera," in *Proc. 21st Korea-Japan Joint Workshop Frontiers Comput. Vis. (FCV)*, Jan. 2015, pp. 1–4.
- [7] D. Zamalieva and A. Yilmaz, "Background subtraction for the moving camera: A geometric approach," *Comput. Vis. Image Understand.*, vol. 127, pp. 73–85, Oct. 2014.
- [8] M.-N. Chapel and T. Bouwmans, "Moving objects detection with a moving camera: A comprehensive review," *Comput. Sci. Rev.*, vol. 38, Nov. 2020, Art. no. 100310, doi: [10.1016/j.cosrev.2020.100310](#).
- [9] Y. Yu, L. Kurnianggoro, and K.-H. Jo, "Moving object detection for a moving camera based on global motion compensation and adaptive background model," *Int. J. Control, Autom. Syst.*, vol. 17, pp. 1866–1874, Jul. 2019, doi: [10.1007/s12555-018-0234-3](#).
- [10] H. Sajid, S.-C.-S. Cheung, and N. Jacobs, "Motion and appearance based background subtraction for freely moving cameras," *Signal Process., Image Commun.*, vol. 75, pp. 11–21, Jul. 2019.
- [11] E. Luna, J. C. San Miguel, D. Ortego, and J. M. Martínez, "Abandoned object detection in video-surveillance: Survey and comparison," *Sensors*, vol. 18, no. 12, p. 4290, Dec. 2018, doi: [10.3390/s18124290](#).
- [12] M. Mandal and S. K. Vipparthi, "An empirical review of deep learning frameworks for change detection: Model design, experimental frameworks, challenges and research needs," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6101–6122, Jul. 2022, doi: [10.1109/TITS.2021.3077883](#).
- [13] E. Komagal and B. Yogameena, "Foreground segmentation with PTZ camera: A survey," *Multimedia Tools Appl.*, vol. 77, no. 17, pp. 22489–22542, Sep. 2018, doi: [10.1007/s11042-018-6104-4](#).
- [14] H. Kong, J.-Y. Audibert, and J. Ponce, "Detecting abandoned objects with a moving camera," *IEEE Trans. Image Process.*, vol. 19, no. 8, pp. 2201–2210, Aug. 2010, doi: [10.1109/TIP.2010.2045714](#).
- [15] M. Fischler and R. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, pp. 381–395, Jun. 1981.
- [16] G. H. F. de Carvalho, L. A. Thomaz, A. F. da Silva, E. A. B. da Silva, and S. L. Netto, "Anomaly detection with a moving camera using multiscale video analysis," *Multidimensional Syst. Signal Process.*, vol. 30, pp. 311–352, Apr. 2019.
- [17] H. Mukojima, D. Deguchi, Y. Kawanishi, I. Ide, H. Murase, M. Ukai, N. Nagamine, and R. Nakasone, "Moving camera background-subtraction for obstacle detection on railway tracks," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2016, pp. 3967–3971, doi: [10.1109/ICIP.2016.7533104](#).
- [18] L. A. Thomaz, E. Jardim, A. F. da Silva, E. A. B. da Silva, S. L. Netto, and H. Krim, "Anomaly detection in moving-camera video sequences using principal subspace analysis," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 3, pp. 1003–1015, Mar. 2018, doi: [10.1109/TCSI.2017.2758379](#).
- [19] E. Jardim, L. A. Thomaz, E. A. B. da Silva, and S. L. Netto, "Domain-transformable sparse representation for anomaly detection in moving-camera videos," *IEEE Trans. Image Process.*, vol. 29, pp. 1329–1343, 2020, doi: [10.1109/TIP.2019.2940686](#).
- [20] R. Padilla, A. F. da Silva, E. A. B. da Silva, and S. L. Netto, "Change detection in moving-camera videos with limited samples using twin-CNN features and learnable morphological operations," *Signal Process., Image Commun.*, vol. 115, Jul. 2023, Art. no. 116969.
- [21] M. K. Panda, B. N. Subudhi, T. Veerakumar, and V. Jakhetiya, "Modified ResNet-152 network with hybrid pyramidal pooling for local change detection," *IEEE Trans. Artif. Intell.*, vol. 5, no. 4, pp. 1599–1612, Apr. 2024.
- [22] U. Panigrahi, M. K. Panda, P. Kumar, S. R. Parija, D. K. Rout, and A. K. Samantaray, "Moving object detection for thermal video using encoder-decoder type deep learning framework," in *Proc. Parul Int. Conf. Eng. Technol. (PICET)*, May 2024, pp. 1–6.
- [23] U. Panigrahi, P. K. Sahoo, M. K. Panda, and G. Panda, "A resnet-101 deep learning framework induced transfer learning strategy for moving object detection," *Image Vision Comput.*, vol. 146, Jun. 2024, Art. no. 105021.
- [24] M. K. Panda, A. Sharma, V. Bajpai, B. N. Subudhi, V. Thangaraj, and V. Jakhetiya, "Encoder and decoder network with resnet-50 and global average feature pooling for local change detection," *Comput. Vision Image Understand.*, vol. 222, Apr. 2022, Art. no. 103501.
- [25] P. K. Sahoo, M. K. Panda, U. Panigrahi, G. Panda, P. Jain, M. S. Islam, and M. T. Islam, "An improved VGG-19 network induced enhanced feature pooling for precise moving object detection in complex video scenes," *IEEE Access*, vol. 12, pp. 45847–45864, 2024, doi: [10.1109/ACCESS.2024.3381612](#).
- [26] M. K. Panda, B. N. Subudhi, T. Veerakumar, V. Jakhetiya, and T. Bouwmans, "A multi-scale contrast preserving encoder-decoder architecture for local change detection from thermal video scenes," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 7968–7981, 2024, doi: [10.1109/TIFS.2024.3447237](#).
- [27] M. K. Panda, B. N. Subudhi, T. Bouwmans, V. Jakhetiya, and T. Veerakumar, "An end to end encoder-decoder network with multi-scale feature pulling for detecting local changes from video scene," in *Proc. 18th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Nov. 2022, pp. 1–8.
- [28] B. N. Subudhi, M. K. Panda, T. Veerakumar, V. Jakhetiya, and S. Esakkirajan, "Kernel-induced possibilistic fuzzy associate background subtraction for video scene," *IEEE Trans. Computat. Social Syst.*, vol. 10, no. 6, pp. 1314–1325, Jun. 2023.
- [29] F. Khozeimeh, D. Sharifrazi, N. H. Izadi, J. H. Joloudari, A. Shoeibi, R. Alizadehsani, M. Tartibi, S. Hussain, Z. A. Sani, M. Khodatars, D. Sadeghi, A. Khosravi, S. Nahavandi, R.-S. Tan, U. R. Acharya, and S. M. S. Islam, "RF-CNN-F: Random forest with convolutional neural network features for coronary artery disease diagnosis based on cardiac magnetic resonance," *Sci. Reports*, vol. 12, p. 11178, Jul. 2022, doi: [10.1038/s41598-022-15374-5](#).

- [30] S. Agajanian, O. Oluyemi, and G. Verkhivker, "Integration of random forest classifiers and deep convolutional neural networks for classification and biomolecular modeling of cancer driver mutations," *Frontiers Mol. Bioscience*, vol. 6, no. 44, pp. 1–17, Jun. 2019, doi: [10.3389/fmolb.2019.00044](https://doi.org/10.3389/fmolb.2019.00044).
- [31] W. Bailer and H. Fassold, "Few-shot object detection using online random forests," in *Proc. Joint Austrian Comput. Vis. Robot. Workshop*, 2020, pp. 95–97, doi: [10.3217/978-3-85125-752-6-22](https://doi.org/10.3217/978-3-85125-752-6-22).
- [32] X. Shen, F. Darmon, A. A. Efros, and M. Aubry, "Ransac-flow: Generic two-stage image alignment," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2020, pp. 618–637.
- [33] V. Krishna, A. Joshi, D. Vrabac, P. Bulterys, E. Yang, S. Fernandez-Pol, A. Y. Ng, and P. Rajpurkar, "GloFlow: Whole slide image stitching from video using optical flow and global image alignment," in *Proc. Int. Conf. Med. Image Comput. Comput. Assisted Intervent.*, 2021, pp. 519–528.
- [34] S.-H. Lai and M. Fang, "A hybrid image alignment system for fast and precise pattern localization," *Real-Time Imag.*, vol. 8, no. 1, pp. 22–23, Feb. 2002, doi: [10.1006/rtim.2001.0256](https://doi.org/10.1006/rtim.2001.0256).
- [35] W. Wen, W. Ren, Y. Shi, Y. Nie, J. Zhang, and X. Cao, "Video super-resolution via a spatio-temporal alignment network," *IEEE Trans. Image Process.*, vol. 31, pp. 1761–1773, 2022, doi: [10.1109/TIP.2022.3146625](https://doi.org/10.1109/TIP.2022.3146625).
- [36] Y. Tian, Y. Zhang, Y. Fu, and C. Xu, "Tdan: Temporally-deformable alignment network for video super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit.*, Jun. 2020, pp. 3360–3369.
- [37] A. F. da Silva, L. A. Thomaz, G. Carvalho, M. T. Nakahata, E. Jardim, J. F. L. de Oliveira, E. A. B. da Silva, S. L. Netto, G. Freitas, and R. R. Costa, "An annotated video database for abandoned-object detection in a cluttered environment," in *Proc. Int. Telecommun. Symp. (ITS)*, Sao Paulo, Brazil, Aug. 2014, pp. 1–5.
- [38] (2020). *Video Database of Abandoned Objects (VDAO)*. Accessed: Sep. 6, 2023. [Online]. Available: https://www02.smt.ufrj.br/~tvdigital/database/objects/page_01.html
- [39] (2020). *Video Database of Abandoned Objects With Annotated Silhouette*. Accessed: Sep. 6, 2023. [Online]. Available: https://www02.smt.ufrj.br/~tvdigital/database/objects/page_02.html
- [40] (2013). *Detection Dataset*. Accessed: Apr. 13, 2025. [Online]. Available: <http://www.svcl.ucsd.edu/projects/anomaly/dataset.htm>
- [41] N. Goyette, P.-M. Jodoin, F. Porikli, J. Konrad, and P. Ishwar, "Changede-tection.Net: A new change detection benchmark dataset," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2012, pp. 1–8, doi: [10.1109/CVPRW.2012.6238919](https://doi.org/10.1109/CVPRW.2012.6238919).
- [42] T. L. B. Dias, L. G. C. Tavares, R. Padilla, A. F. da Silva, L. A. Thomaz, S. L. Netto, and E. A. B. da Silva, "Siamese networks for bounding-box to silhouette annotation of video databases," in *Proc. Simpósio Brasileiro de Telecomunicações e Processamento de Sinais*, Jan. 2020, pp. 1–21, doi: [10.14209/SBRT.2020.1570645009](https://doi.org/10.14209/SBRT.2020.1570645009).
- [43] J. Sánchez Pérez, E. Meinhardt-Llopis, and G. Facciolo, "TV-L1 optical flow estimation," *Image Process. Line*, vol. 3, pp. 137–150, Jul. 2013, doi: [10.5201/ipol.2013.26](https://doi.org/10.5201/ipol.2013.26).
- [44] D. Chicco, *Siamese Neural Networks: An Overview*. Cham, Switzerland: Springer, 2020, doi: [10.1007/978-1-0716-0826-5](https://doi.org/10.1007/978-1-0716-0826-5).
- [45] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Kyoto, Japan, Jun. 2009, pp. 248–255, doi: [10.1109/CVPR.2009.5206848](https://doi.org/10.1109/CVPR.2009.5206848).
- [46] B. M. Afonso, L. P. Cinelli, L. A. Thomaz, A. F. da Silva, E. A. B. da Silva, and S. L. Netto, "Moving-camera video surveillance in cluttered environments using deep features," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 2296–2300, doi: [10.1109/ICIP.2018.8451540](https://doi.org/10.1109/ICIP.2018.8451540).
- [47] T. Liu, A. Abd-Elrahman, J. Morton, and V. L. Wilhelm, "Comparing fully convolutional networks, random forest, support vector machine, and patch-based deep convolutional neural networks for object-based wetland mapping using images from small unmanned aircraft system," *GIScience Remote Sensing*, vol. 55, pp. 243–264, Jul. 2018, doi: [10.1080/15481603.2018.1426091](https://doi.org/10.1080/15481603.2018.1426091).
- [48] (2023). *Light GBMs Documentation*. Accessed: Oct. 10, 2023. [Online]. Available: <https://lightgbm.readthedocs.io/en/stable/index.html>
- [49] F. Archetti and A. Candelier, *Bayesian Optimization and Data Science*. Cham, Switzerland: Springer, 2019.
- [50] (2020). *Scikit-optimize: Sequential Model-based Optimization in Python*. Accessed: Jan. 24, 2024. [Online]. Available: http://scikit-optimize.github.io/stable/auto_examples/bayesian-optimization.html
- [51] (2020). *200-frame Excerpts From VDAO Database Used for Testing Algorithms*. Accessed: Sep. 6, 2023. [Online]. Available: http://www02.smt.ufrj.br/~tvdigital/database/research/page_01.html
- [52] B. W. Matthews, "Comparison of the predicted and observed secondary structure of T4 phage lysozyme," *Biochimica et Biophysica Acta*, vol. 405, no. 2, pp. 442–451, Oct. 1975, doi: [10.1016/0005-2795\(75\)90109-9](https://doi.org/10.1016/0005-2795(75)90109-9).
- [53] S. Boughorbel, F. Jarray, and M. El-Anbari, "Optimal classifier for imbalanced data using Matthews correlation coefficient metric," *PLoS ONE*, vol. 12, Jun. 2017, Art. no. e0177678.
- [54] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 1, pp. 1–13, Dec. 2020.
- [55] D. Chicco, N. Tötsch, and G. Jurman, "The Matthews correlation coefficient (MCC) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation," *BioData Mining*, vol. 14, no. 1, pp. 1–22, Feb. 2021.



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